MATH 2101 Linear Algebra I, FALL 2025

Tutorial 9 Solution

1. The possible values of a are 0, $-2/3$ because

$$1 = \left\| \begin{pmatrix} a \\ a+1 \\ -a \end{pmatrix} \right\|^2 = a^2 + (a+1)^2 + (-a)^2 = 3a^2 + 2a + 1,$$

which is equivalent to $a(3a+2) = 0$, that is, $a = 0$ or $a = -\frac{2}{3}$.

2. Let $w = (x_1, x_2, x_3, x_4)^T$. We suffice to solve the linear system

$$\begin{cases} x_1 + x_2 + 3x_3 + 5x_4 & = 0 \\ 3x_1 + 2x_2 + x_3 & = 0 \end{cases}.$$

$\vdots$

The general solution is $w = (5s + 10t, -8s - 15t, s, t)^T$ where $s, t \in \mathbb{R}$.

3. Note that $v = \frac{\langle u_1, v\rangle}{\|u_1\|^2}u_1 + \frac{\langle u_2, v\rangle}{\|u_2\|^2}u_2 + \frac{\langle u_3, v\rangle}{\|u_3\|^2}u_3 = \cdots = -\frac{1}{3}u_1 + \frac{5}{3}u_2 - u_3$. So $[v]_\beta = (-1/3,\ 5/3,\ -1)^T$.

4. Let $u = (2,\ 3,\ 4)^T$. The orthogonal projection of $v = (1,\ 1,\ 1)^T$ onto W is

$$\frac{\langle u, v\rangle}{\|u\|^2}u = \frac{2+3+4}{2^2+3^2+4^2}\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \frac{9}{29}\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}.$$

5. (a) Disproof: Take $u_i = \mathbf{e}_i$ for $i = 1, 2, 3$, then $S^\perp = \operatorname{span}(\{\mathbf{e}_4\})$. So

$$S \cap S^\perp = \varnothing$$

is not a subspace of $\mathbb{R}^4$.

(b) Proof: As $S^\perp$ is a subspace of $\mathbb{R}^4$, we have $S^\perp \cap (S^\perp)^\perp = \{0\}$, which is a subspace of $\mathbb{R}^4$.

(c) Proof: Note that $\operatorname{span}(S) \neq \mathbb{R}^4$. Fix $w \in \mathbb{R}^4 \backslash \operatorname{span}(S)$ and take

$$v = w - \langle u_1, w\rangle u_1 - \langle u_2, w\rangle u_2 - \langle u_3, w\rangle u_3 \in S^\perp.$$

Then $v \neq 0$, otherwise $w \in \operatorname{span}(S)$ which contradicts that $w \in \mathbb{R}^4 \backslash \operatorname{span}(S)$. Hence $\{u_1, u_2, u_3, v\}$ is an orthogonal set of non-zero vectors, and hence linearly independent.

(d) Proof: Take $v = 0 \in S^\perp$, then $\{u_1, u_2, u_3, v\} = \{u_1, u_2, u_3, 0\}$ is linearly dependent.